

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Text Structure and Purpose	Medium

Question

Using NASA’s powerful James Webb Space Telescope (JWST), Mercedes López-Morales and colleagues measured the wavelengths of light traveling through the atmosphere of WASP-39b, an exoplanet, or planet outside our solar system. Different molecules absorb different wavelengths of light, and the wavelength measurements showed the presence of carbon dioxide (CO₂) in WASP-39b’s atmosphere. This finding not only offers the first decisive evidence of CO₂ in the atmosphere of an exoplanet but also illustrates the potential for future scientific breakthroughs held by the JWST.

Which choice best describes the overall structure of the text?

Answer

- A. It discusses a method used by some researchers, then states why an alternative method is superior to it.
- B. It describes how researchers made a scientific discovery, then explains the importance of that discovery.
- C. It outlines the steps taken in a scientific study, then presents a hypothesis based on that study.
- D. It examines how a group of scientists reached a conclusion, then shows how other scientists have challenged that conclusion.

Correct Answer: B

Rationale

Choice B is the best answer. The text begins by describing how the researchers used the JWST to detect CO₂ in WASP-39b’s atmosphere. Then the text discusses the significance of this finding, both as the first evidence of CO₂ in an exoplanet’s atmosphere and as an illustration of the JWST’s potential for making new discoveries in general.

Choice A is incorrect. The text doesn’t compare two different methods, but rather focuses on one study that used the JWST. Choice C is incorrect. The text doesn’t present a hypothesis, but rather reports on the findings of a study. Choice D is incorrect. The text doesn’t mention any scientists challenging the conclusion reached by López-Morales and colleagues.